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New Insights into $T-H/H-F$ Diagrams for Synthesis of Heat Exchanger Networks inside Heat Integrated Water Allocation Networks

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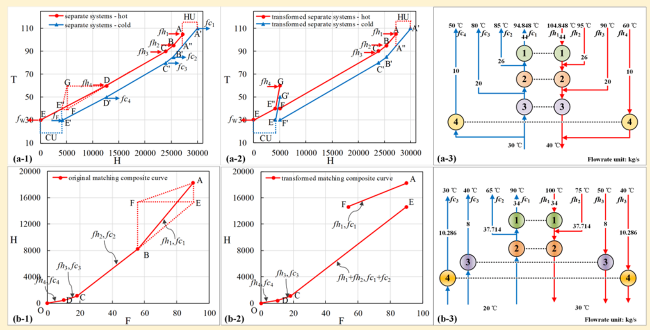
Supporting Information

ABSTRACT: Energy and resources saving is of significance to sustainable development. In the past decades, heat integrated water allocation networks (HIWANs) have been applied to generate effective strategies for water and energy management in chemical process industries, where large amounts of water and energy are consumed. In this paper, a graphical approach for the synthesis of heat exchanger networks (HENs) in HIWANs has been proposed to reveal the new insights into $T-H/H-F$ diagrams and fully exploit the HEN structure possibilities. Current conceptual design methods for HENs in HIWANs by the original separate systems ($T-H$ diagram) and the matching composite curve ($H-F$ diagram) only generate two special kinds of HEN structures: series and parallel structures. A transformation method is proposed to generate HENs with different structures based on these two tools. Both simplex series or parallel structures and hybrid structures can be obtained via the transformation procedures. Two examples are illustrated to demonstrate the operability of the proposed procedures.



■ INTRODUCTION

Process integration has been recognized as a powerful framework to achieve sustainable design of industrial processes.[1] Water integration and energy integration, related to the synthesis of water allocation networks[2] (WANs) and heat exchanger networks[3] (HENs), are two key categories of process integration, especially for chemical process industries, which consume vast amounts of water and energy. Since water and energy are inextricably intertwined in process industries, the synthesis of heat integrated water allocation networks (HIWANs) has drawn more attention, owing to the simultaneous and efficient utilization of water and energy. Two recent comprehensive reviews have been contributed by Ahmetovic et al.[4] and Kermani et al.[5]

Two main categories for the synthesis of HIWANs can be found: conceptual design and mathematical programming. Conceptual design methods are generally based on physical insights and heuristics, while the mathematical models are based on the superstructures, such as stagewise superstructure based models,[6,7] trans-shipment type HEN models,[8,9] state-space representation models,[10] and separate wastewater treatment models.[11] The mathematical programming methods can be classed into sequential and simultaneous approaches. The simultaneous approaches are proficient in considering trade-offs between different factors, such as the freshwater consumption, the utility consumption, and the capital cost of heat exchangers and wastewater treatment units.[12,13] However, it is still challenging to obtain a global optimal solution or even a good local optimum, especially for complex large-scale problems.[14,15] Although the conceptual design method is sequential in nature, which leads to suboptimal solutions, it does not need sophisticated solution schemes. Besides, it has the advantages of process insights. Despite the limitations, conceptual design methods perform well in generating promising solutions, as well as good initial points for mathematical programming models.

The first conceptual design method for the synthesis of HIWANs was contributed by the research group in Manchester.[16] The WAN and HEN were designed by the new grid representation of two-dimensional grid diagram and the separate systems approach based on the composite curve in the $T-H$ diagram, respectively. Two different situations, no water reuse[17] and maximum water reuse,[18] were investigated.

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Subsequently, Leewongtanawit and Kim[19] developed a Water Energy Balance Diagram to improve energy recovery in the WAN, followed by a new separate systems generation approach to minimize heat exchanger areas. Liao et al.[20] developed a graphical tool based on $H-F$ diagrams, which was applied to get a HEN with parallel structures. Additionally, conceptual design methods have been developed for the problems with mass transfer-based and nonmass transfer-based water-using operations, such as Heat Surplus Diagram by Manan et al.[21] Later, Wan Alwi et al.[22] extended this approach by superimposing source and demand allocation curves and Heat Surplus Diagram on a plot of flow rate versus mass load/temperature, termed as Superimposed Mass and Energy Curves (SMEC), while six design heuristics were introduced to guide design toward the minimum utility targets. In addition to the approaches for the synthesis problems with single contaminant, Hou et al.[23] adopted the concentration potential concept[24] to deal with multiple contaminants problems, presented by a newly developed Temperature and Concentration Order Composite Curves (TCOCC). In their subsequent research,[25] the method was extended for nonmass transfer-based systems with both single and multiple contaminants, while a single-temperature-peak design principle was proposed to guide the synthesis procedures. Besides, Luo et al.[26,27] explored the effect of different types of nonisothermal mixing on the utility target. Some mixing rules were applied to simplify the HEN structure and avoid the penalty caused by improper nonisothermal mixing. Recently, Hong et al.[28] developed a novel graphical tool, namely Heat Transfer Blocks Diagram, to target utility consumption and guide the synthesis of HEN to lower investment cost direction.

The WAN and HEN are sequentially designed in conceptual design methods. It can be seen that most of above-mentioned conceptual design methods for the synthesis of HIWANs focus on the synthesis of WANs, in order to simultaneously reduce freshwater and utility consumption. The synthesis of HENs mainly relies on the $T-H$ diagram and the $H-F$ diagram, resulting in HENs with series structures and parallel structures, respectively. The design of series structures mainly depends on the $T-H$ diagram and the separate systems proposed by Savulescu et al.,[17,18] while the design of parallel structures are obtained by the $H-F$ diagram and the matching composite curve of Liao et al.[20] However, these methods only obtain HENs with simplex series or parallel structures. Recently, we obtained series−parallel hybrid structures via novel graphical tools in the T−F diagram.[28] As a result, the design space for minimum utility consumption have been enlarged greatly. Nevertheless, it is desired to obtain hybrid HEN structures in the $T-H$ and $H-F$ diagrams, because they are essential in analyzing a HEN inside a HIWAN. This is the object of this paper. In this paper, new insights of the $T-H$ and $H-F$ diagrams are revealed and more HEN structure possibilities can be obtained by $T-H$ and $H-F$ diagrams. A transformation method is developed to generate HENs with different structures based on the original separate systems and the matching composite curve. Hybrid structures can be obtained via the proposed method. Two examples are presented to illustrate the operability of the proposed procedures.

New Insights of $T-H$/$H-F$ Diagrams. Let us introduce and compare the $T-H$ diagram and the $H-F$ diagram first. Table 1 compares the $T-H$ diagram and the $H-F$ diagram in the following aspects: the abscissa, the ordinate, and the slope. A $T-H$ diagram and an $H-F$ diagram are shown in Figure 1-a-1 and b-1, respectively. For simplicity, it is assumed that there is no hot utility and no cold utility needed. The hot/cold composite curves in both Figure 1-a-1 and b-1 are marked by red/blue dashed lines, respectively. Besides, the separated systems are also presented in Figure 1-a-1 by red and blue solid lines, while the matching composite curve is presented in Figure 1-b-1 by a black solid line. The separate systems and the matching composite curves can be obtained by mixing and splitting streams, as shown in Figure 1-a-2 and b-2 respectively. The detailed procedures can be found in the work of Savulescu et al.[17,18] and Liao et al.,[20] respectively. Based on the separate systems and the matching composite curve, the corresponding HENs can be obtained, as shown in Figure 1-a-2 and b-2.

Table 1. Comparisons between $T-H$ and $H-F$ Diagrams

diagram	abscissa	ordinate	slope
$T-H$	enthalpy	temperature	flow rate
$H-F$	flow rate	enthalpy	temperature difference

The significant characteristic of these two methods is that both the line sections of the separate systems in the $T-H$ diagram and the matching composite curve in the $H-F$ diagram stand for heat exchanger matches. However, the involved hot/cold streams behind these two kinds of line sections are different. Each line section of the separate systems represents a set of hot/cold streams in the certain temperature interval. For example, line section CD represent the mixing stream of hot streams fh_1, fh_2, and fh_3. Since hot stream fh_1 covers all three temperature intervals, it exists at line sections AB, BC, and CD in the $T-H$ diagram of Figure 1-a-1. Thus, hot stream fh_1 flows from point A to point D in Figure 1-a-2, going through heat exchangers 1, 2, and 3. As a result, hot streams fh_1, fh_2, and fh_3 go through three, two, and one heat exchanger(s) in series, respectively. Nevertheless, each line section represents a hot/cold stream in its whole temperature range in the $H-F$ diagram. Thus, each hot (cold) stream goes through only one heat exchanger, cooled down (heated up) from its starting temperature to its ending temperature, as shown Figure 1-b-2, where the heat exchangers are arranged in parallel.

In conclusion, both the separate systems and the matching composite curves are constructed to find heat exchange matches, based on temperature intervals and streams, respectively. It means that all streams in the same temperature interval are mixed by separate systems, while each stream is individually managed in matching composite curves. Accordingly, we can get different HENs by transforming the separate systems and the matching composite curve. The first step is to find the line sections representing the specified heat exchangers. The second step is to decompose the involved line sections. The third step is to assemble the decomposed line sections. The last step is to obtain the new HEN structure based on the transformed matching composite curve or separate systems. Note that, the following discussion is based on hot and cold streams of the separate systems and the matching composite curve, not the original hot and cold composite curves. In the following sections, two examples[18,29] are illustrated to introduce the methodology on the $H-F$ diagram and $T-H$ diagram, respectively. The operation data of two examples is listed in Tables S1 and S2 of the Supporting Information.

Transformation on the $H-F$ Diagram. A well-known example involving four water-using operations given by

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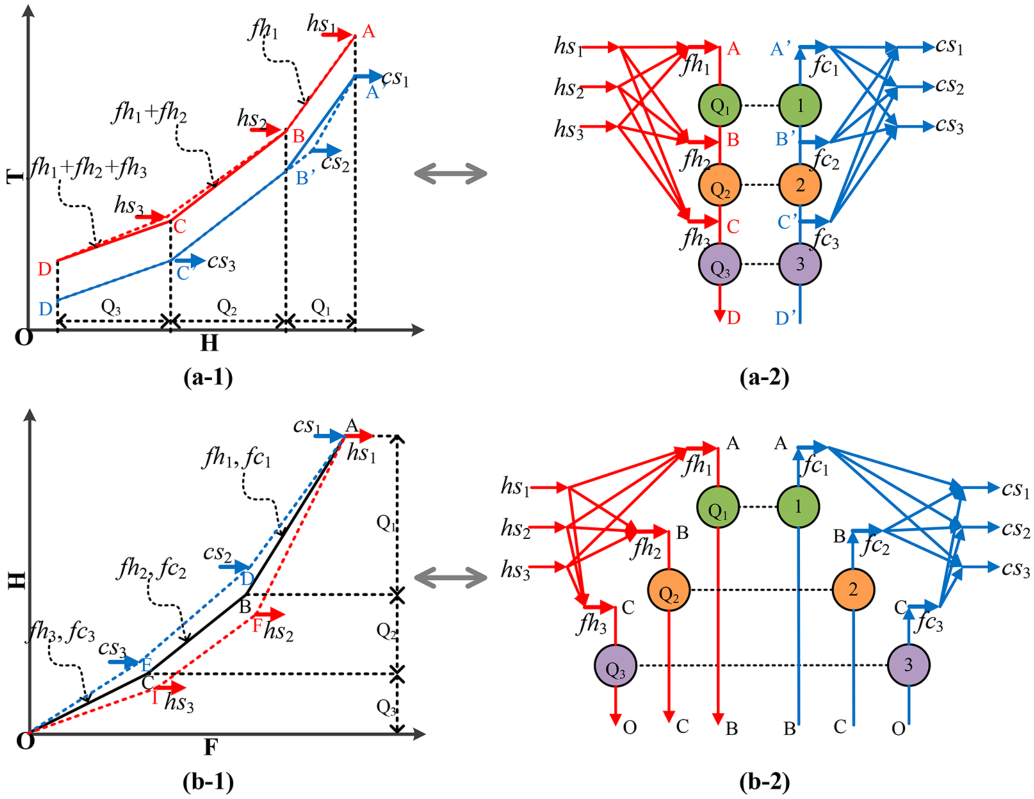

Figure 1. Illustration for the separate systems on the $T-H$ diagram and the matching composite curve on the $H-F$ diagram.

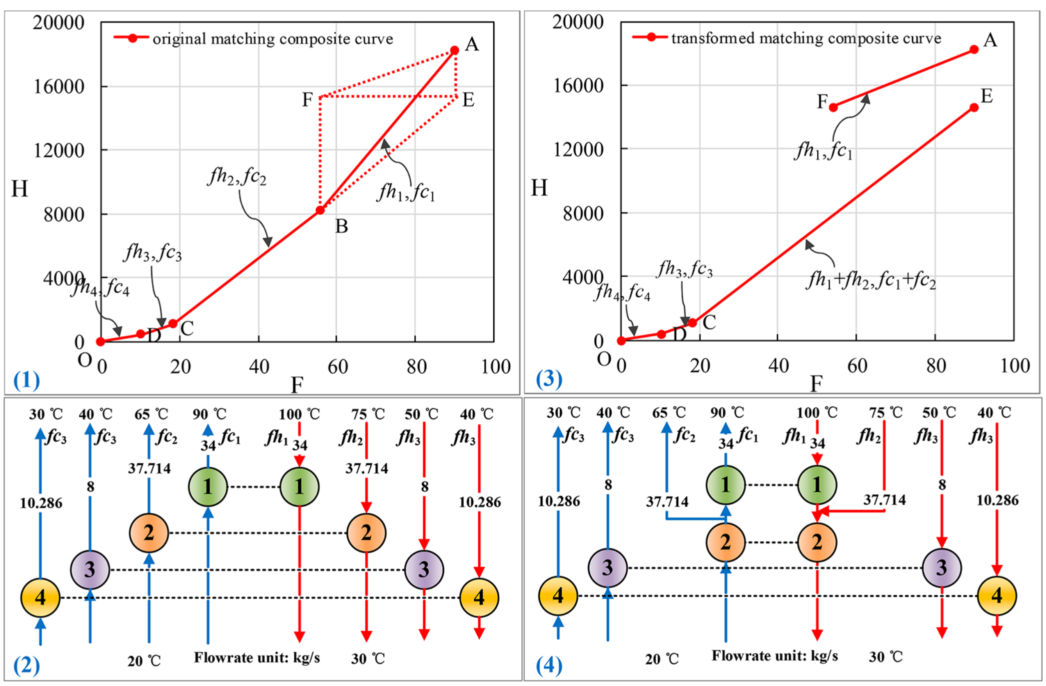

Figure 2. Transformation procedures of the matching composite curve.

Savulescu and Smith[17] is illustrated on the $H-F$ diagram, based on the result of our previous work.[20] In our previous work, a matching composite curve with four line sections was generated, as shown in Figure 2-1, resulting in a HEN with four parallel heat exchangers shown in Figure 2-2. The purpose is to integrate heat exchangers 1 and 2. Since the manipulation of hot streams and cold streams are similar, we only discuss the transformation of the matching composite curve on hot streams for simplicity.

Step 1: Find the line sections representing the specified heat exchangers. In this example, heat exchangers 1 and 2 are represented by line sections AB and BC.

Step 2: Decompose the involved sections. Let us extend line section CB to intersect with the vertical line through point A at

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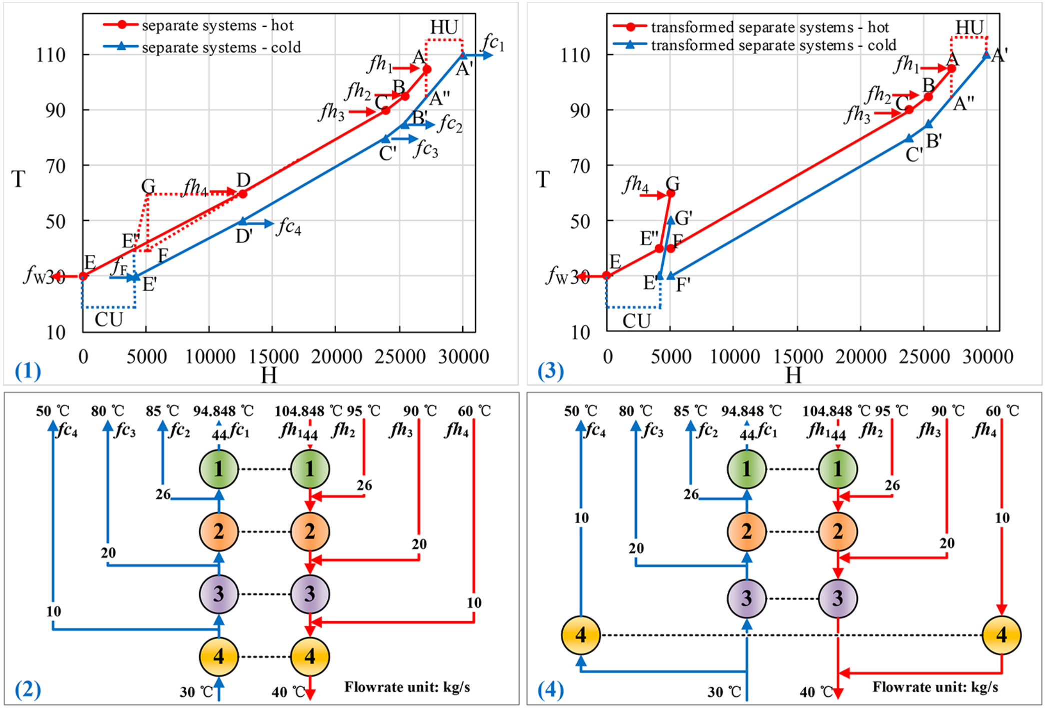

Figure 3. Transformation procedures of the separate systems.

point E, and then draw a horizontal line through point E and a vertical line through point B. The obtained horizontal line and vertical line will be intersected at point F, as shown in Figure 2-1. Then, connect point A and point F. So far, line section AB is divided into two line sections AF and BE.

Step 3: Assemble the decomposed line sections. Since the slope of the line section EB is equal to the one of line section BC, line section EB represents hot stream fh_1 in the temperature interval of 75−30 °C. Thus, line sections EB and BC are assembled to line section EC, which represents the mixing stream of hot streams fh_1 and fh_2 in the temperature interval of 75−30 °C. According to energy balance, line section AF must represent stream fh_1 in the temperature interval of 100−75 °C.

Step 4: Obtain the new HEN structure based on the transformed matching composite curve. Four line sections AF, EC, CD, and DO in Figure 2-3 correspond to four heat exchangers with heat loads of 3570, 13 554, 672, and 432 kW in Figure 2-4, respectively. The overall HIWAN is shown in Figure S1 of the Supporting Information.

Since the temperature differences for all heat exchangers are given, the sum of heat exchanger areas also remains unchanged during the transformation procedures. However, the heat load and the area of each heat exchanger will be redistributed. According to the heuristic in our previous research,[28] constructing larger heat exchanger will decrease the capital cost. After the transformation procedures from Figure 2-2 to 2-4, the heat load of the heat exchangers 1 and 2 change from 9996 and 7128 kW to 3570 and 13554 kW, respectively. A heat exchanger with a larger heat load of 13554 kW is constructed. The capital costs of these two exchangers change from 122,735 and 101,665 \$/y to 69,859 and 145,733 \$/y, respectively, resulting a total capital cost decrease of 8808 \$/y. Under the heuristic,[28] the transformation procedures not only construct HENs with different structures but also could decrease the total capital cost of heat exchanger networks.

Transformation on the *T*−*H* Diagram. The transformation on the $T-H$ diagram is illustrated on a large-scale example[29] consisting of 15 water-using operations. The separate systems can be generated according to the water allocation network from Liu et al.,[29] as shown in Figure 3-1, while the corresponding HEN with series structures is shown in Figure 3-2. Note that, hot utility and cold utility are not presented in Figure 3-2, thus, the presented streams are energy balanced. It is assumed that the target is separate hot stream fh_4 from the series structures, so that hot stream fh_4 and the mixing stream of hot streams fh_1, fh_2, and fh_3 are distributed in parallel in the temperature interval of 60−40 °C. The transforming procedures are illustrated in Figure 3-1. Note that, only the transformation of hot composite curve is illustrated in Figure 3-1, while only hot streams are marked in Figure 3-3.

Step 1: Find the line sections representing the specified heat exchangers. It is known that the series to parallel structure change happens on heat exchanger 4. Thus, the involved line section of Figure 3-1 is line section DE″.

Step 2: Decompose the involved sections. The target is to separate hot stream fh_4 from line section DE″. Let us extend line section CD, intersected with a horizontal line through point E″ at point F, as shown in Figure 3-1. Then, draw a vertical line through point F and a horizontal line through point D, intersected at point G. Next, connect point E″ and point G. So far, line section DE″ is divided into two line sections DF and E″G, which represents the mixing stream of hot streams fh_1, fh_2, and fh_3 and hot stream fh_4 in the temperature interval of 60−40 °C, respectively.

Step 3: Assemble the decomposed line sections. Since line section CD and DF represent the mixing stream of hot streams fh_1, fh_2, and fh_3 in the temperature intervals of 90−60 °C and 60−40 °C, respectively, the line section CF must represent the mixing stream of hot streams fh_1 and fh_2 in the temperature intervals of 90−40 °C.

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Step 4: Obtain the new HEN structure based on the transformed matching composite curve. According to the final transformed separate systems in Figure 3-3, hot stream fh_4 is cooled down from its starting temperature 60 to 40 °C, then cooled down to its ending temperature 30 °C together with hot streams fh_1, fh_2, and fh_3, as shown in Figure 3-4. The overall HIWAN is shown in Figure S2 of the Supporting Information.

Similar to the previous example, the heat load of heat exchangers are redistributed after the transformation from Figure 3-2 to 3-4. The heat load of heat exchangers 3 and 4 change from 11340 and 8400 kW to 18900 and 840 kW, respectively. The new heat exchanger 3 with the larger heat load of 18900 kW is generated, resulting in the total capital cost decrease of 33,014 $/y.

■ CONCLUSION

This paper addresses the synthesis of HENs in HIWANs based on new insights into the $T-H$ and $H-F$ diagrams. A transformation method is developed to identify HENs with different structures via several steps: first identify the involved line sections, then decompose and reassemble them, and finally obtain the new HEN structures. Owing to the advantage of the visualization, the heat load of heat exchangers, the flow rate of hot/cold streams, and the temperature of hot/cold streams can be acquired easily from the diagrams. Compared to the previous methods, the proposed approach with new insights can obtain more kinds of HENs, simplex series or parallel structures, and series−parallel hybrid structures. This process highly improves the structure freedom of HENs obtained by the $H-F$ and $T-H$ diagrams. Using the proposed approach, it allows decision-makers to screen and scope promising networks with different structures. Besides, with the heuristic,[28] the transformation could decrease the total capital cost of heat exchanger networks. Two examples show different HEN structures that satisfy the minimum utility consumption can be obtained by the transformation procedures.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.8b00978.

Operation data of examples 1 and 2; HIWAN for examples 1 and 2 (PDF)

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Notes

The authors declare no competing financial interest.

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